

BRIGHT LEARN EXERCISE 1: SQL FUNDAMENTALS: SELECT & FILTERING

1 SELECTING & SORTING

employees

id	first_name	last_name	department	Salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

01 SELECT *
FROM employees;

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

02 SELECT DISTINCT
FROM employees;

department
IT
HR
Finance
Marketing

```

03 SELECT first_name,
        last_name
FROM   employees
ORDER BY salary DESC;

```

first_name	last_name	salary
Eva	Naidoo	72000
Alice	Green	70000
Cathy	Zulu	65000
David	Mokcena	50000
Brian	Lee	45000

```

04 SELECT id,
        first_name,
        last_name,
        salary
FROM   employees
ORDER BY salary DESC
LIMIT 3;

```

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

205 FILTERING WITH WHERE

```

05 SELECT id,
        first_name,
        last_name,
        department
FROM   employees
WHERE  department = 'IT';

```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

06 SELECT id,
first_name,
last_name,
department,
salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;

id	first_name	last_name	department	salary
03	Cathy	Zulu	Finance	65000

07 SELECT id,
first_name,
last_name,
department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08 SELECT id,
first_name,
last_name,
department
FROM employees
WHERE department NOT IN ('IT');

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokone	Marketing

09 SELECT id,
first_name,
last_name,
department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10 SELECT id,
first_name,
last_name,
department,
salary,
city
FROM employees
WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg